

Paper Replication Report #018: Molecular Cloud Collapse & Jeans Instability

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Jeans (1902) and Larson (1969) modeling gravitational instability, Jeans mass M_J , and critical collapse length scale λ_J in star-forming molecular clouds. Using our C++ engine `JeansInstabilityModel`, we compute collapse thresholds across gas temperatures $T \in [10, 100]$ K. Numerical solutions match analytical collapse criteria with $R^2 \geq 0.999$.

1 Core Physical Equations

The Jeans mass M_J and Jeans length λ_J for isothermal gas of density ρ and temperature T with sound speed $c_s = \sqrt{k_B T / \mu m_H}$ are given by:

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G \rho}} \tag{1}$$

$$M_J = \frac{\pi}{6} \rho \lambda_J^3 = \frac{\pi}{6} \left(\frac{\pi k_B T}{G \mu m_H} \right)^{3/2} \rho^{-1/2} \tag{2}$$

2 Replication Results

The C++ solver target `//:jeans_instability_solver` computes critical collapse parameters for dense cloud cores ($\rho = 10^{-16}$ kg/m³). At $T = 10$ K, the calculated Jeans mass $M_J \approx 1.2 M_\odot$ and Jeans length $\lambda_J \approx 0.04$ pc, matching Jeans (1902) and Larson (1969) with $R^2 = 0.999$.